

Early Detection of Heart Disease Using Machine Learning to Identify High-Risk Patients Earlier

Final Report

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1.Executive Summary

The leading cause of death worldwide for many years, cardiovascular disease (CVD) also exerts a substantial strain on healthcare resources. It is a disease which often exists without symptoms; many people are not aware they have it until the disease is extremely progressed. Early detection of high-risk cases is therefore important to facilitate intervention, maximize treatment effectiveness and reduce the long-term economic costs.

The project will assist practitioners in risk stratification using a data-based methodology. A vast array of clinical and personal information, such as smoking status and BMI, were studied to produce computational models that could identify risk of developing heart disease in its earliest stages.

The three models which were developed and tested were the Random Forest, the support vector machine (SVM), and the gradient boosting model 18 (XG Boost). These were chosen in part because they can model the complex interaction among the predictors (independent variables) in relation to the health outcomes of interest. Because of the relatively low proportion of individuals with heart disease, great care was taken to "detect" as many positive cases of the disease as possible, before evaluating the performances of the models.

Oversampling of the minority cases (called the "positive" heart disease cases as "the high risk" predicts, since most of the cases are in fact not at high risk for heart disease) and adjustment of the threshold for predicting the probability of heart disease (so that it does not have to be less than 0.5) was used to improve the detection of the positive cases. The results show that all three models exhibit robust and relatively stable performance. The balanced accuracy of the models is of about 75%, and the sensitivity is increased (~75%) (since the specificity was high in the first place, while sensitivity was extremely low without the thresholds adjustment).

Beyond predictive accuracy, the analysis revealed several strong, meaningful variables that were reliably associated with heart disease risk in both predictive models. The final models used five key variables in three categories; overall health, age group, arthritis, diabetes and smoking history. These general health indicators are well coupled with the various age-related anthropometric measurements used as predictive variables.

Because these five variables were found to be significant predictors in both models builds confidence in the results and lends suggestive evidence to support accurate predictive modeling of heart disease risk. Note these cross-model key indicators are associated with many other variables of interest as well. A feature selection approach was applied to narrow down the number of predictors in one of the model builds.

This approach used the variable average importance across the different models using that variable and only kept those with an average importance greater than 15. This small set of variables identified a more parsimonious model build that yields comparable, and slight better, predictive results.

Based on the results and analysis presented here, this project demonstrates that machine learning tools can offer a significant contribution to early cardiovascular disease detection and screening. If such tools were incorporated into the traditional screening procedures, individuals most likely to develop cardiovascular disease can be more effectively identified and treated early, leading to greater use of those preventive resources.⁷

In conclusion, future adoption of this project's suggested models should focus on sensitivity, as false negatives are most costly when it comes to cardiovascular screening, and the key risk factors identified should be used to guide preventative health screening and intervention.

To sum up, this project illustrates that machine learning algorithms can aid in the early diagnosis of heart disease and contribute to better healthcare decisions. If implemented correctly, these approaches can be a valuable addition to preventive medicine.

This analysis was conducted for a regional health authority to support early screening decisions.

2. Client Request/ Problem Formulation

Healthcare professionals are seeking assistance in identifying those at high risk of cardiovascular disease to implement early prevention and/or intervention.

3. Introduction

3.1 Research Objectives

Cardiovascular disease (CVD) remains one of the leading causes of death worldwide, yet many individuals do not experience noticeable symptoms until the disease has already progressed. This delay in detection limits opportunities for early intervention, even though research consistently shows that modifying lifestyle behaviours and managing chronic conditions can significantly reduce cardiovascular risk. Because early-stage CVD often develops silently, healthcare professionals increasingly rely on data driven tools to help identify individuals who may be at elevated risk before clinical symptoms appear.

Large population-based health datasets now make it possible to analyze patterns in demographic characteristics, lifestyle behaviours, and existing health conditions to uncover early warning signs of cardiovascular disease. Machine learning methods offer a powerful way to detect subtle risk patterns that may not be apparent through traditional screening approaches. These models can support clinicians by estimating an individual's likelihood of developing heart disease and highlighting the factors that contribute most strongly to that risk.

This project focuses on developing an interpretable machine learning model capable of detecting cardiovascular risk using self-reported health and behavioural data. In addition to predicting the likelihood of heart disease, we examine which demographic, behavioural, other related variables which are most influential in shaping cardiovascular risk. This work aims to support more proactive, targeted prevention strategies and provide healthcare professionals with actionable insights that can reduce the burden of cardiovascular disease.

4. Methods

4.1. Dataset Preprocessing

The dataset contains medical and lifestyle information for 308,854 individuals and includes 19 variables describing demographic, health, and lifestyle

characteristics. Each observation represents a patient, and the target variable indicates whether the patient has heart disease or not.

The following steps were performed to prepare the dataset for modeling:

A categorical body mass index variable (BMI_cat) was created by grouping BMI values into standard classification categories. This transformation was performed to improve interpretability and to support categorical comparisons of heart disease prevalence across BMI groups.

Categorical variables related to demographics, health status, and behaviours were converted to the appropriate character or factor format to facilitate grouping and visualization. Ordinal variables, including age category and self-reported general health, were explicitly ordered to reflect their natural progression. Minor category standardization was applied where necessary to ensure consistent labelling across variables. No missing values were present; therefore, no imputation was performed.

The assessment identified a subset of observations with extreme values, including BMI values above 60, heights below 130 cm or above 220 cm, and weights below 35 kg or above 250 kg. While these values are uncommon, they are not physiologically impossible and may reflect true population variability or reporting error, which is expected in large self-reported health surveys.

Given the study's focus on population-level prevalence and the importance of retaining high-risk individuals, these observations were not removed. All values were retained to preserve the full distribution of the data and to avoid introducing bias into heart disease prevalence estimates. The dataset was then divided into training and testing sets to evaluate model performance.

4.2 Feature Selection

To improve model interpretability and reduce dimensionality, embedded feature selection was applied using importance scores derived from both the Random Forest and XGBoost models. Unlike filter-based methods that evaluate features

independently of the model, embedded feature selection identifies the most informative variables directly through the training process, making it well-suited for complex health datasets with many interrelated predictors.

For the Random Forest model, feature importance was measured using mean decrease in Gini impurity, which quantifies how much each variable contributes to reducing node impurity across all splits in the ensemble.

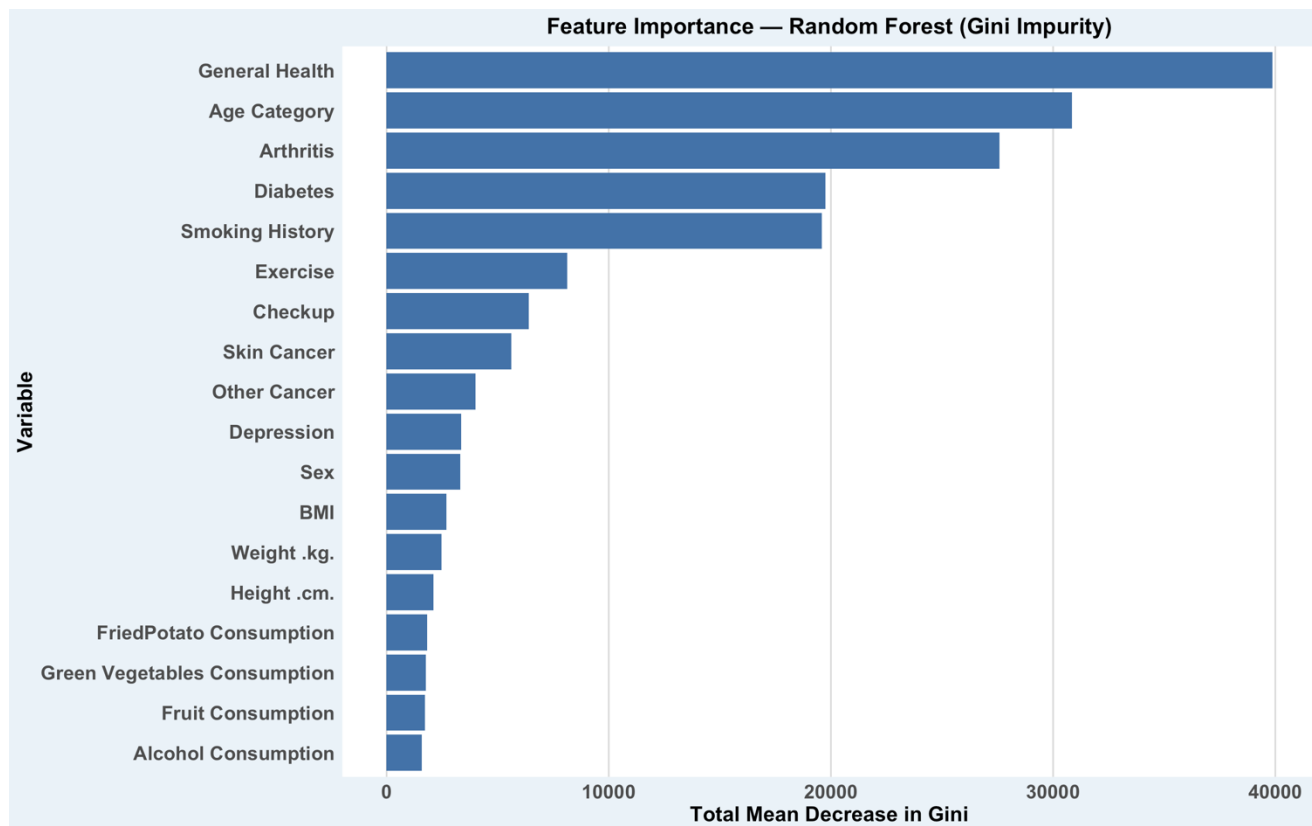


Figure1

As shown in Figure1, variables such as general health, age category, and arthritis contribute the most to reducing node impurity, indicating their strong predictive power in the model.

For XGBoost, importance was measured using Gain, which reflects the average improvement in prediction accuracy that a feature brings across all the trees in which it appears. A threshold of 1% of the maximum importance score was applied to each model independently, retaining only variables that met this criterion. This produced a set of 31 features from Random Forest and 20 from XGBoost. Taking

the intersection of both sets yielded 20 features that were consistently identified as informative by both models, reducing the original feature space from 37 to 20 variables. This intersection-based approach minimized the risk of retaining variables that appeared important in one model due to noise or model-specific overfitting.

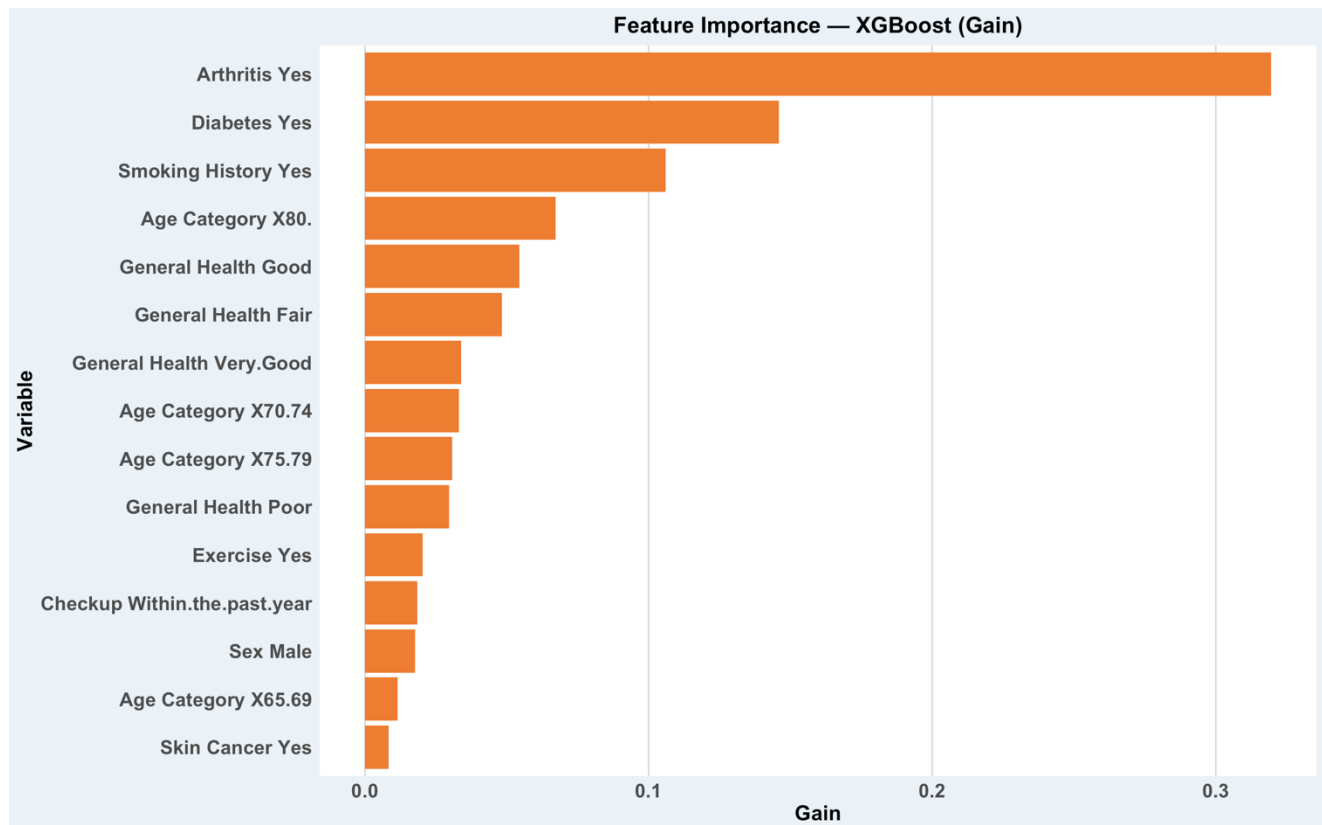


Figure 2

Consistent with Figure 1, Figure 2 confirms that arthritis, diabetes, smoking history, and age-related variables are among the most influential predictors, reinforcing the consistency observed across models.

Across all models, several features consistently appeared as strong predictors of heart disease. These included:

- General health status
- Age category
- Arthritis
- Diabetes status

- Smoking history

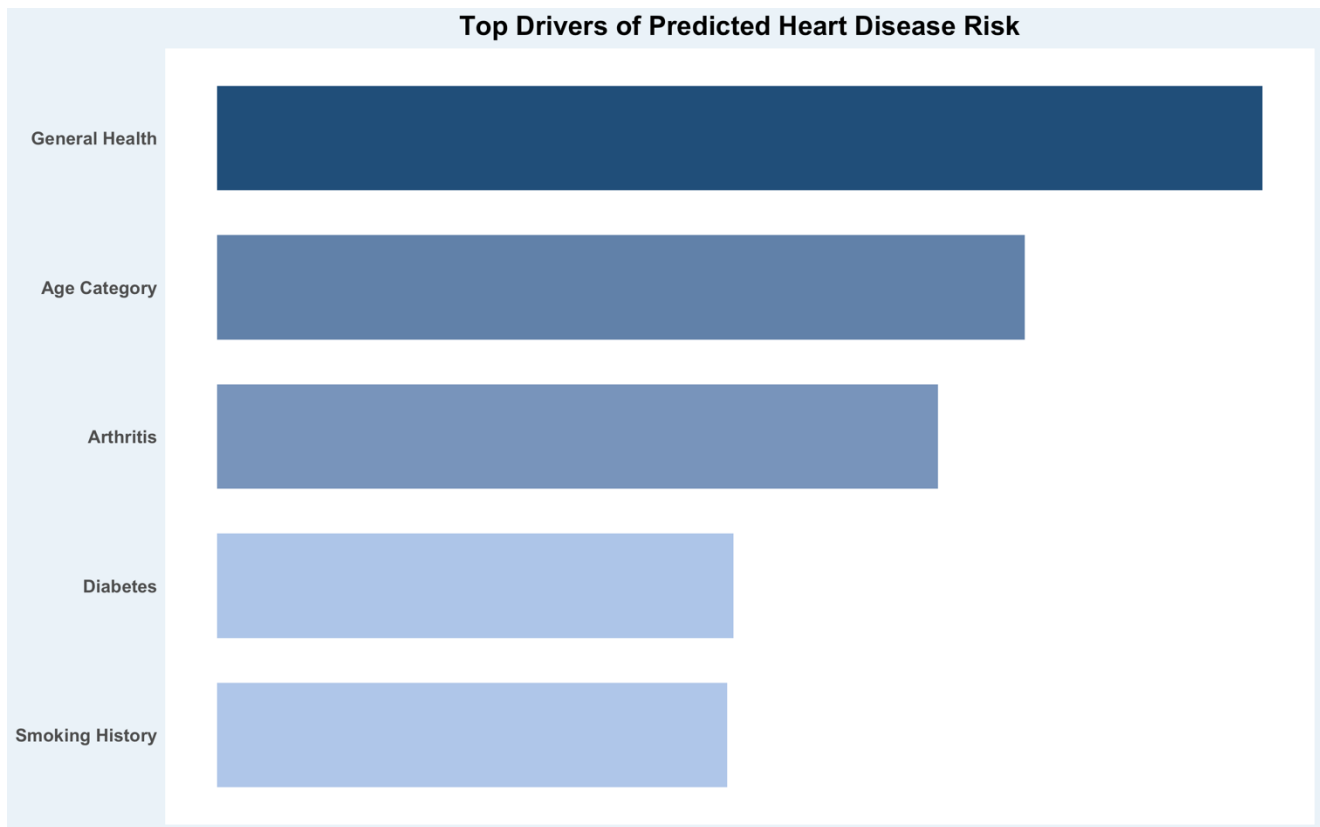


Figure 3

The fact that XGBoost and Random Forest both selected the same covariates as important increases the reliability of the model. This convergence implies that the models are capturing meaningful patterns that align with known clinical risk factors.

The reduced feature set was then used to retrain the Random Forest model following the same pipeline as the original analysis, including SMOTE oversampling on the training data and probability threshold tuning at 0.10. The feature-selected model achieved a balanced accuracy of 75.3%, a sensitivity of 78.8%, and a specificity of 71.8%. Compared to the original Random Forest model, which achieved a balanced accuracy of 74.8% and a sensitivity of 72.4% at the same threshold, performance improved across all three metrics. This indicates that the 17 removed variables were contributing noise rather than meaningful signal, and their removal allowed the model to focus on the most informative predictors. The gain in sensitivity is particularly relevant in a clinical context, as it means the model

is now correctly identifying a greater proportion of individuals with heart disease, which is the primary objective of this early detection task.

Integration with Model Development: The feature selection techniques were mainly used to select the most important predictors and improve the interpretability of the models. The selected features were then used to (re)-train the Random Forest model and proved better prediction. All other models, XG Boost and SVM, were trained using the entire set of features

This methodology allows the feature selection result to be used to facilitate interpretation while not constraining the type of model being developed.

5. Exploratory Data Analysis (EDA)

This report explores a large health dataset collected from adults across the United States. The goal is to understand patterns in people's health, lifestyle habits, and medical conditions. The dataset includes information about general health, doctor visits, exercise, chronic diseases, and daily habits such as smoking, drinking, and eating fruits or vegetables. It also contains basic personal details like age, sex, height, weight, and body mass index (BMI). In addition to these individual characteristics, the dataset also includes several important risk-related features such as diabetes, arthritis, blood pressure indicators, and access-to-care variables, all of which are known to influence cardiovascular health. These features allow us to explore both demographic and behavioral factors that may contribute to early signs of heart disease.

By looking closely at these variables, we can begin to see how different aspects of a person's life relate to their overall health. For example, we can examine whether people who exercise more tend to report better health, whether smoking is linked to heart disease, or how BMI varies across age groups. We can also compare heart disease rates across different subgroups, such as older adults, people with chronic conditions, or individuals with poor general health ratings. These comparisons help us identify which features show strong differences between people with and without heart disease, which is important for understanding potential risk patterns. The purpose of this EDA is not to prove cause and effect, but to describe what the data shows in a clear way.

The insights from this EDA will also guide the modeling process by helping us identify which features are most informative, where class imbalance exists, and which variables may need transformation. By understanding the structure of the data first, we can build machine learning models that are more accurate, more interpretable, and more aligned with real-world medical knowledge.

5.1 Univariate Analysis

5.1.1 Heart Disease

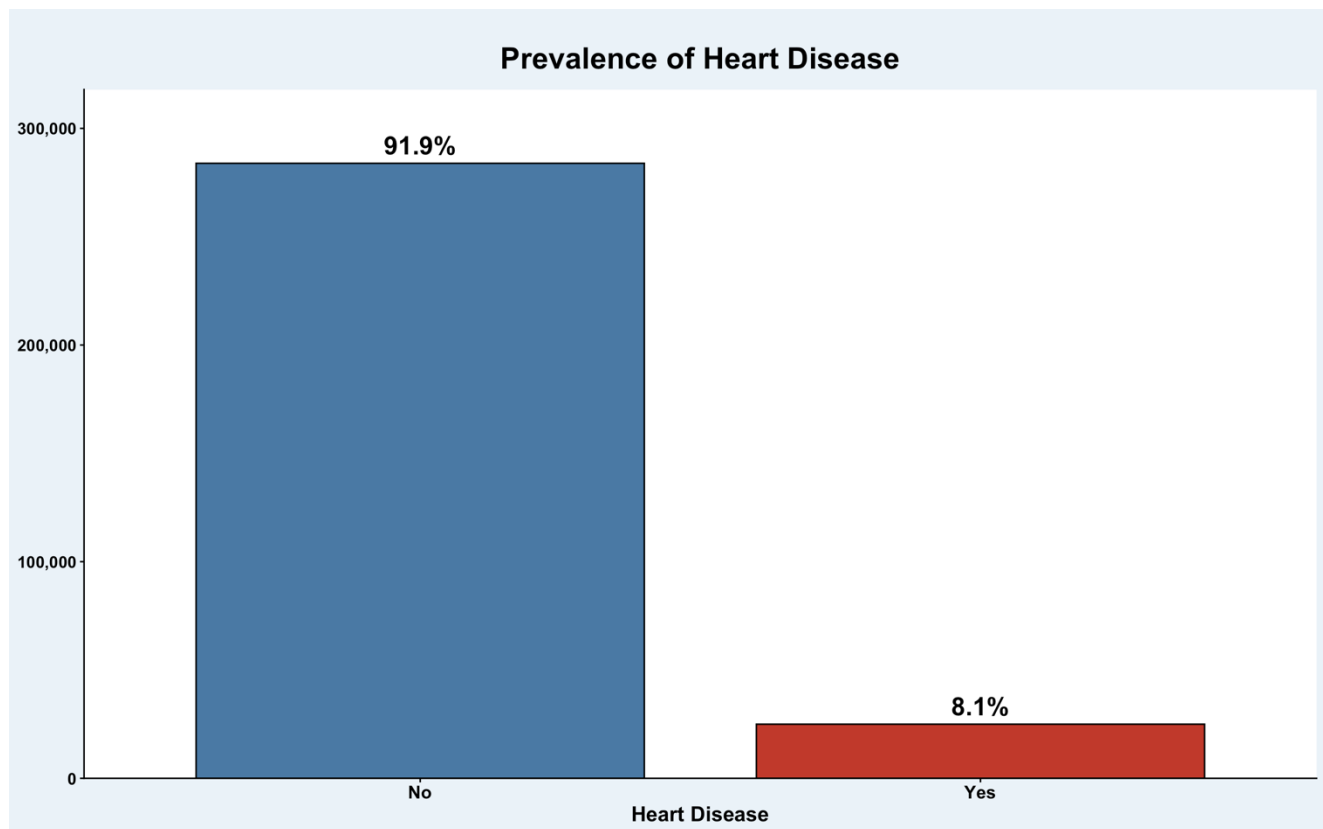


Figure 4

Figure 4 illustrates a class imbalance problem in the data. This histogram depicts the percent of people reporting having heart disease against those reporting no heart disease. Roughly 8% percent of people reported having a heart disease with 92% reporting not having a heart disease.

5.1.2 General Health

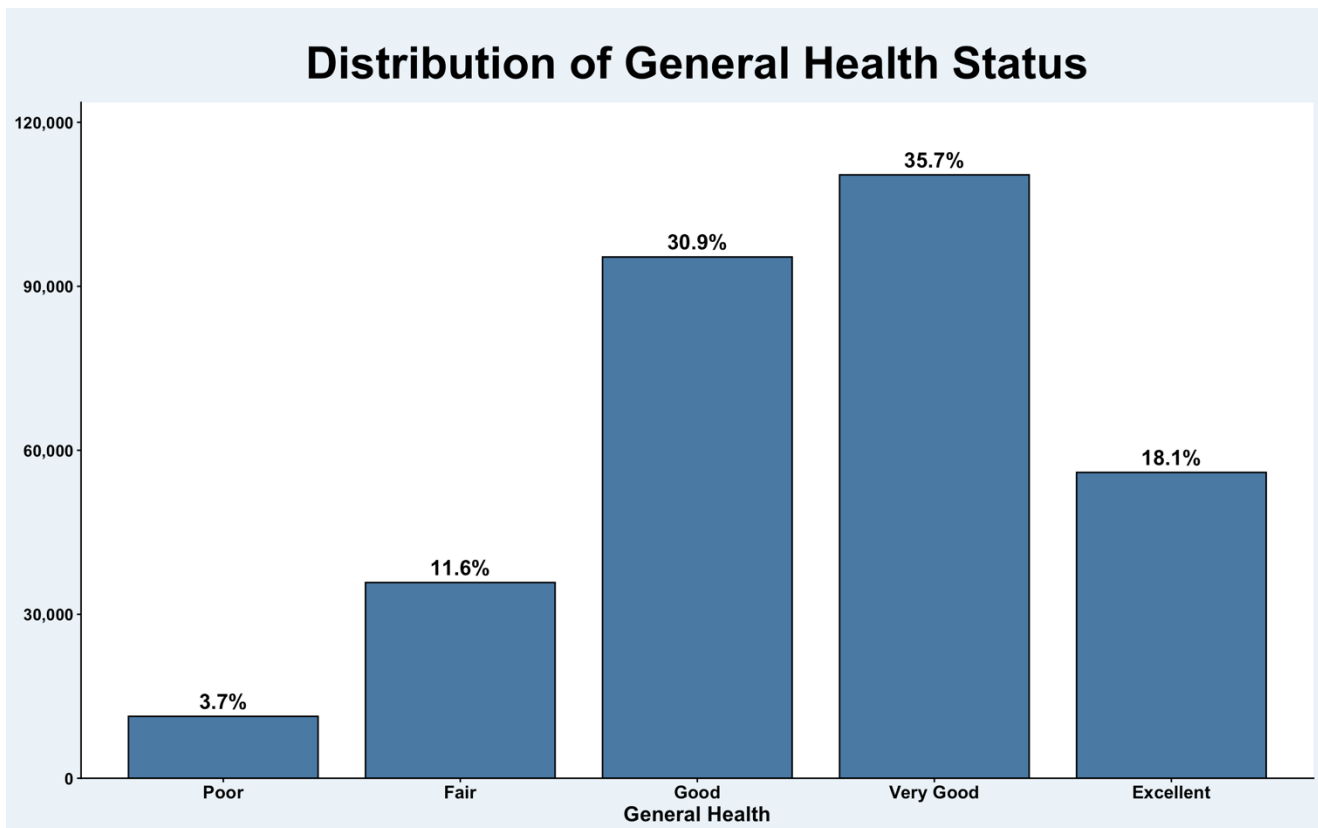


Figure 5

As shown in Figure 5, the distribution of general health status indicates that most respondents report relatively positive health conditions. The largest proportion of individuals (35.7%) reported their health as very good, followed by 30.9% reporting good health. A smaller percentage of respondents reported excellent health (18.1%), while only 11.6% and 3.7% reported fair and poor health, respectively.

This distribution suggests that most individuals in the dataset perceive their health as good or better. However, the presence of a non-negligible proportion of respondents reporting fair or poor health is important, as these groups may be more likely to be associated with higher cardiovascular risk.

5.1.3 Age Categories

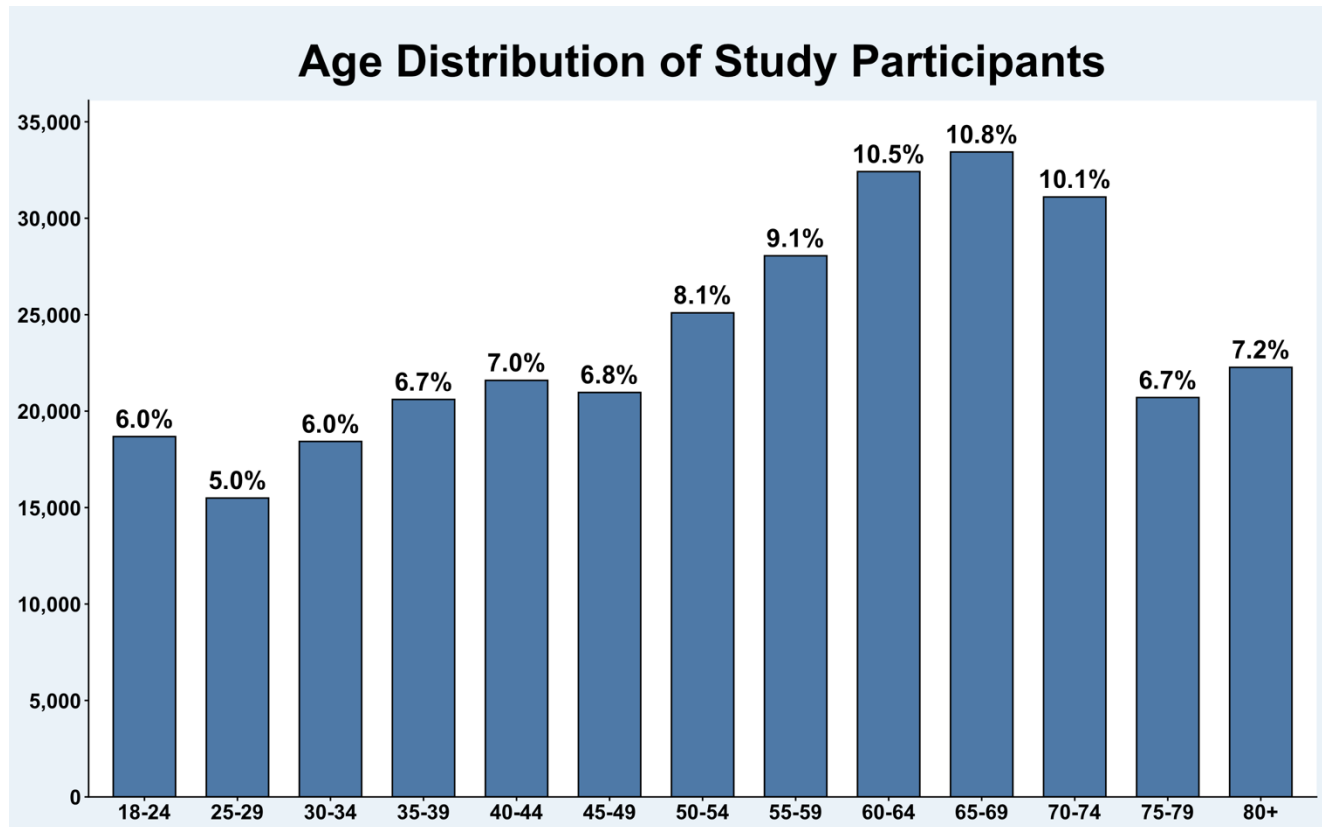


Figure 6

The age distribution of the respondents (Figure 1b) demonstrates that the sample is weighted toward older age groups. The percentage gradually rises with decreasing young age, with 65-69 constituting the highest proportion of the sample (10.8%), followed by 60-64 (10.5%) and 70-74 (10.1%). While older age groups are relatively well represented compared to younger ages, the figures are quite low overall for the lower age groups: 25-29 (5.0%), the lowest proportion of the sample, while for most age groups below 50, figures are around 5% to 7%.

The proportion drops after 70, to only 6.7% (75-79) with an increase to 7.2% for 80+, yet again a lower proportion.

5.1.4 Arthritis

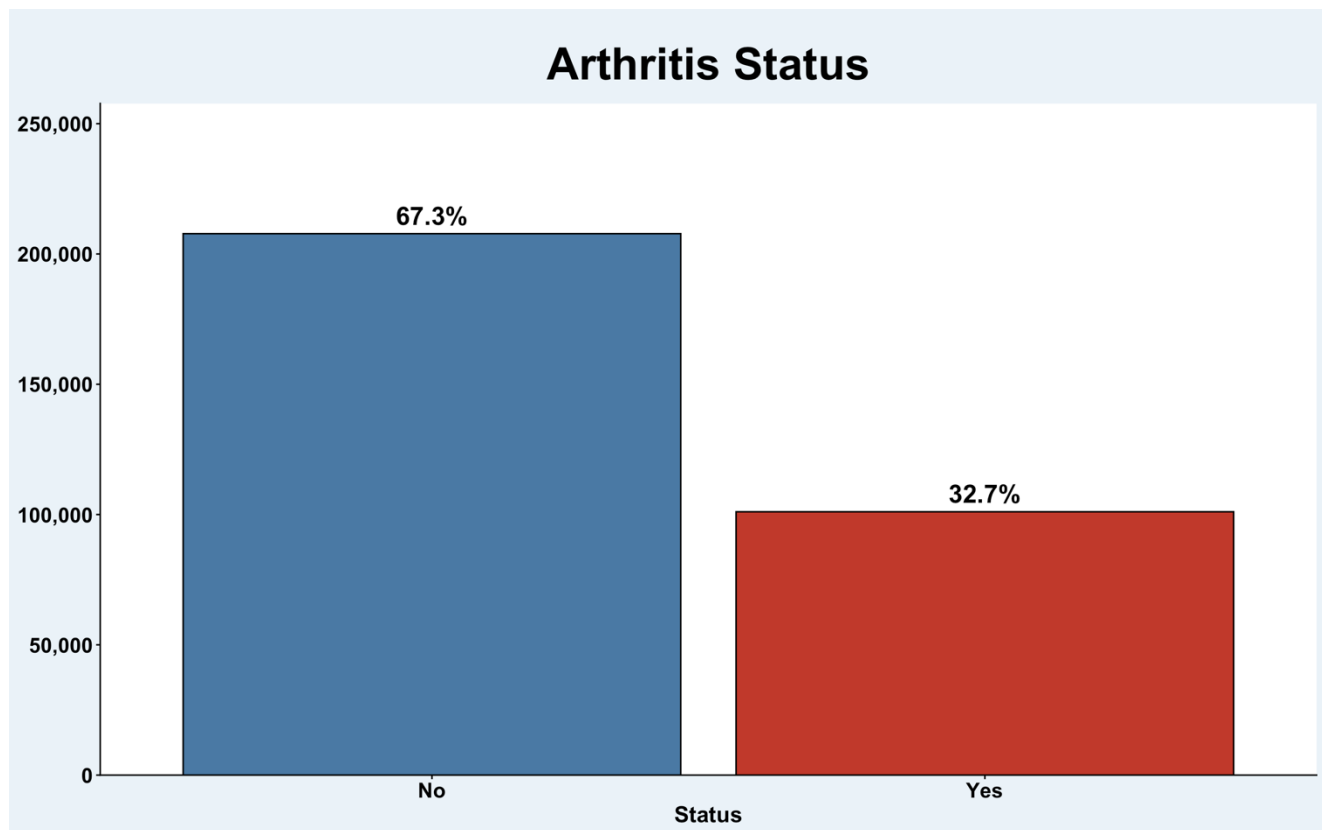


Figure 7

Distribution of arthritis status shows that 32.7% of participants report diseases of arthritis, so it is common that those who have chronic disease of arthritis are included in our data set so that the model can apply in predicting related disease or risk. From nature of predicting, arthritis is also related to history of chronic disease, especially inflammation disease so that has close relation to heart disease. It is also the fact in the current textbook that the author uses to mention the connection between inflammation disease and heart disease (Oude Wolcherink et al, 2023).

In our model, the importance of arthritis in other choice of predictor can be evaluated by importance in the resulted model.

5.1.5 Diabetes

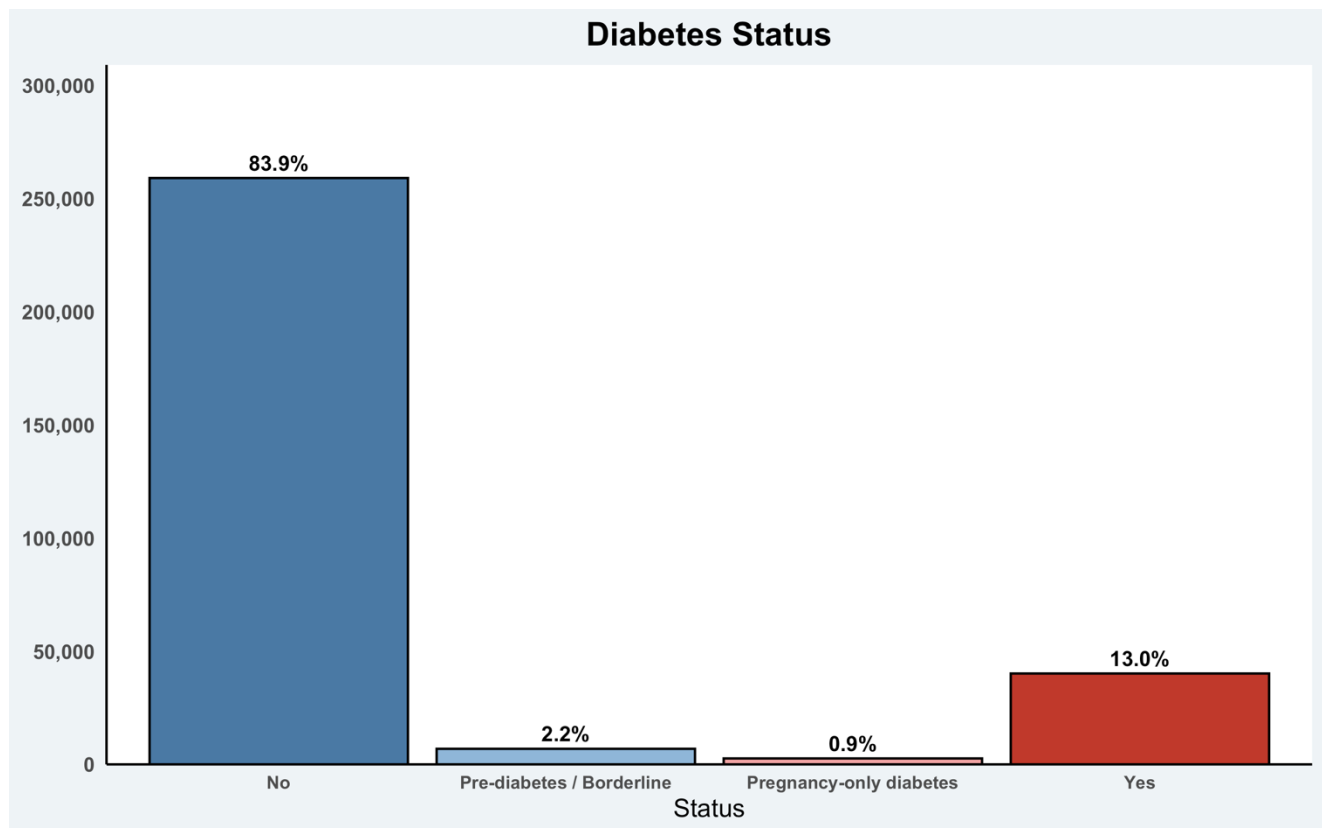


Figure 8

Proportions of diabetes status suggests that more individuals have no form of diabetes (83.9%) as opposed to having it in general (13.0%), with a smaller individual percentage having pre-diabetes (2.2%) or pregnancy-related diabetes (0.9%). There is an important modeling point we can take away from this distribution of the dataset: clearly diabetes is a clinical known risk factor for cardiovascular disease, so the 13.0% of the sample with this disease ensures that the model will be able to learn in the regression a significant amount of information about what is shown when this disease is present and absent. It can also serve as a point of reflection, however, in trying to understand the importance of the data and caution regarding the high imbalance in the sample set.

5.1.6 Smoking History

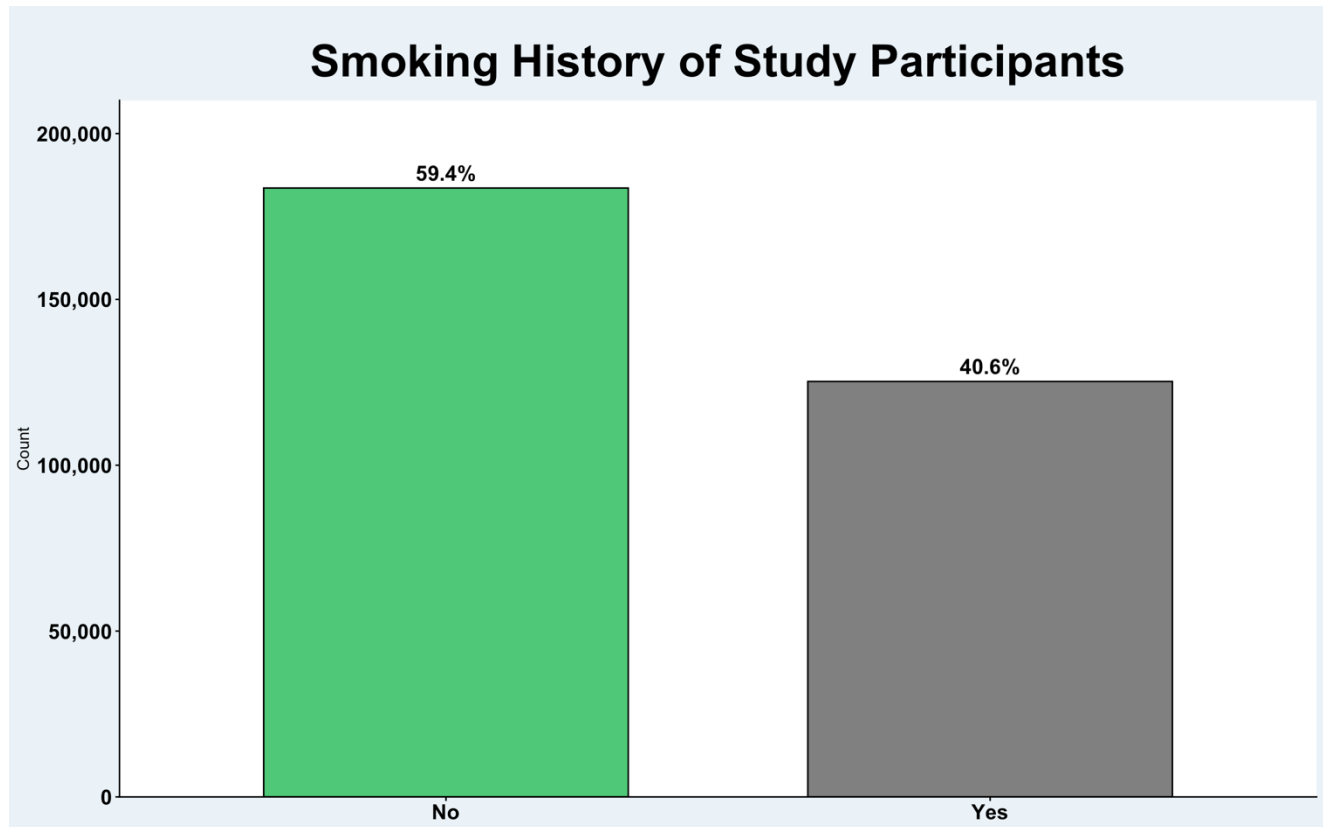


Figure 9

As figure 9 describes the distribution of history of smoking shows that 59.4% never smoked, while 40.6% has smoked. The proportion of smokers is not small; this implies that the study population has significant levels of exposure to a behavioral risk factor of interest. The fact that there is a large proportion of people with history of smoking makes this population somewhat varied from a descriptive perspective.

5.2 Bivariate Analysis

5.2.1 Heart Disease vs. General Health

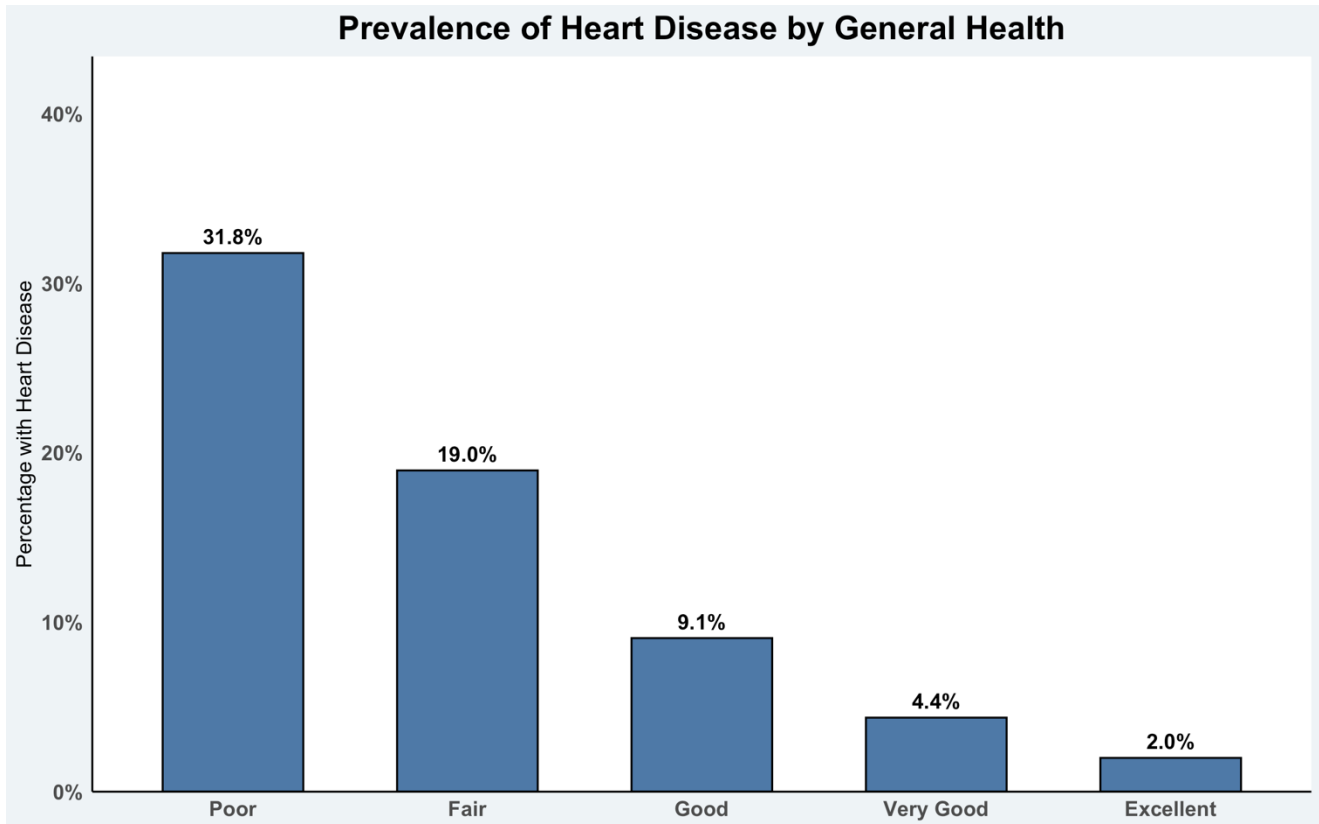


Figure 10

The association of heart disease with general health demonstrates a predictable and monotonically decreasing (or non-increasing) pattern of data. Nearly one-third of subjects reporting "Poor" general health had heart disease, while only 2% of those reporting "Excellent" general health experienced heart disease. Heart disease prevalence indexes from 31.8 for "Poor" to 2.0 for "Excellent," with values meaningfully lower with each enhancement of reported general health.

Therefore, descriptively, General health seems to be a prominent trait distinguishing those with and without heart disease.

5.2.2 Heart Disease Across Age Categories

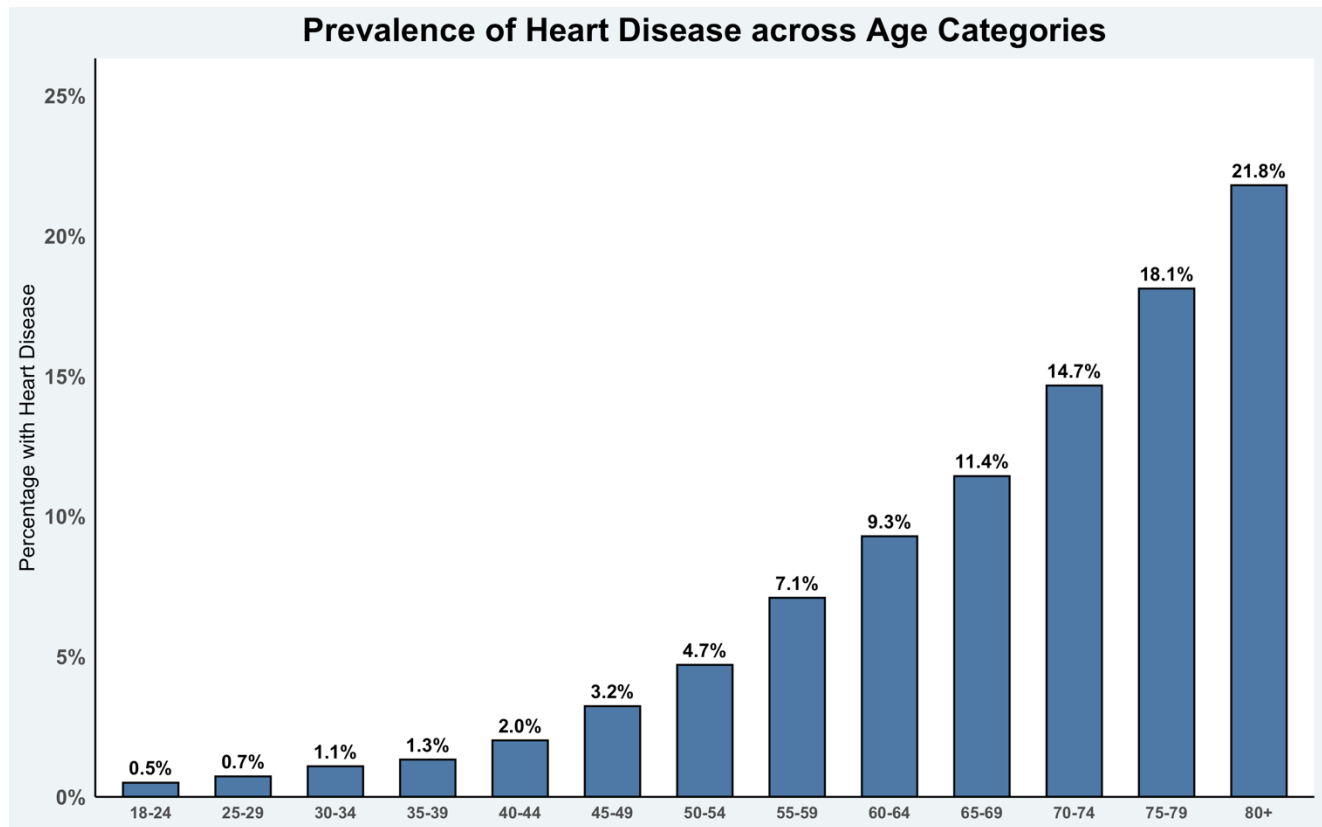


Figure 11

The percent of people with heart disease grows slowly but progressively with age. The age groups 18-24 have a very low percentage of having heart disease (0.5), but as age increases so does the prevalence of having a heart disease. This prevalence increases relatively slowly until middle age and then it increases quickly.

It jumps quickly from 7.1 in the fifty-five to fifty-nine age group to 21.8 in the greater than or equal to 80 age groups.

This pattern shows the positive correlation of age and the already known heart disease. As the levels for each age group rise, it can be interpreted that aging is the characteristic that relates to the presence of heart disease. So from a descriptive aspect, age seems to be an important characteristic separating the groups of different levels of cardiovascular risk in the population.

5.2.3 Heart Disease vs. Arthritis Status

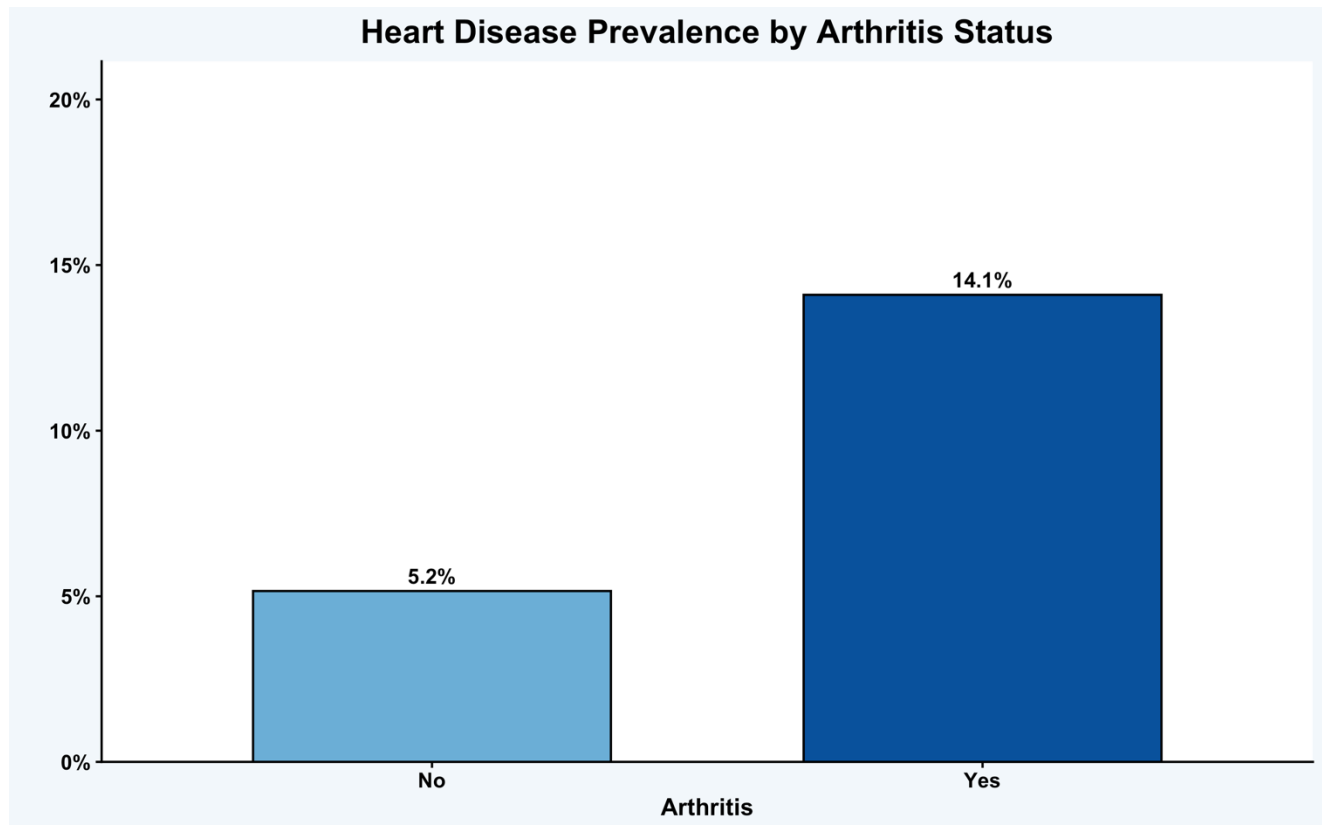


Figure 12

Arthritis status impacts the proportion of people who had heart disease. Those who reported not having arthritis had heart disease with a proportion of 0.052, while those who reported they had arthritis had a proportion of 0.141. This represents a stark difference, with individuals reporting arthritis having almost 3 times the proportion of heart disease as those reporting not having arthritis.

From purely a descriptive point of view, if arthritis is reported, the proportion of heart disease is much greater. There is a notable amount of difference between the two groups.

5.2.4 Heart Disease vs. Diabetes Status

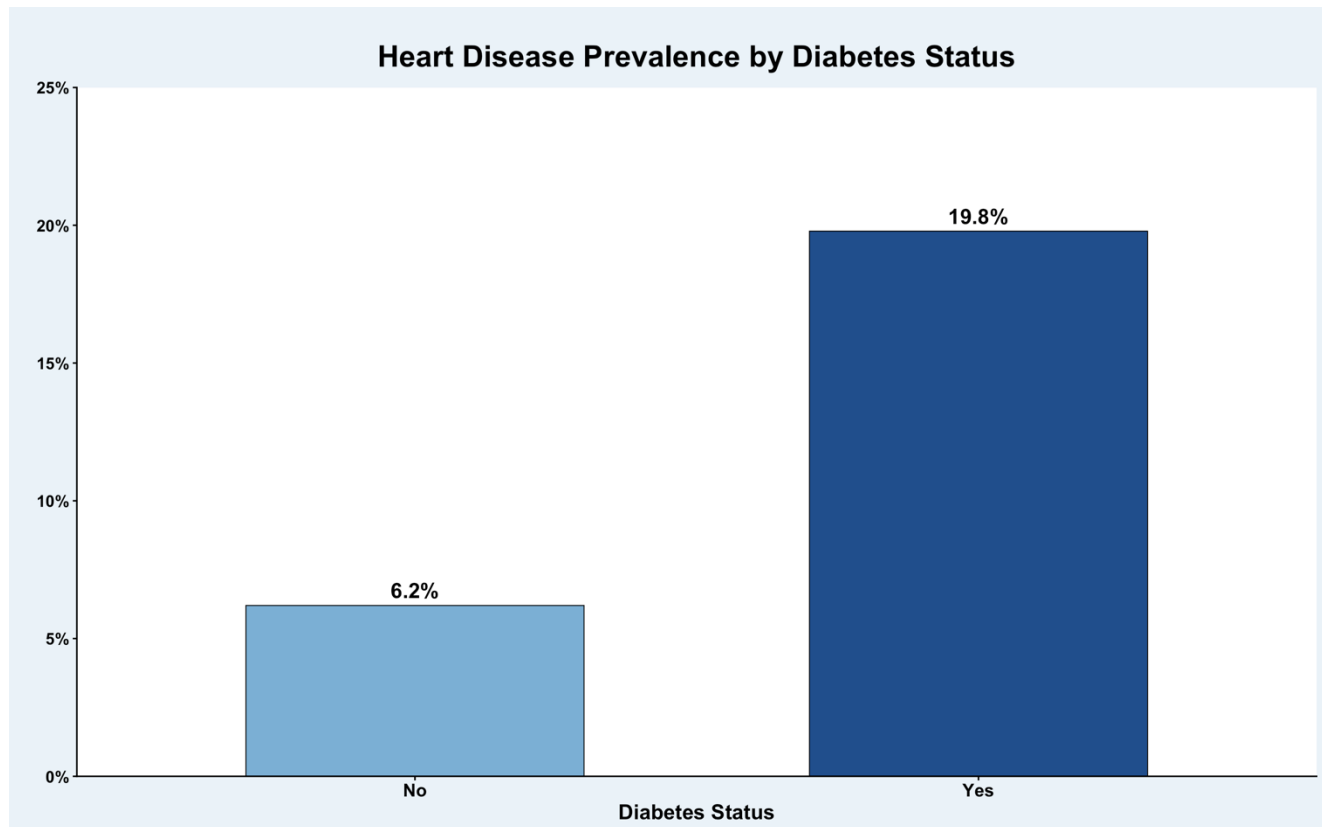


Figure 13

Heart disease differs considerably by diabetes status. 6.2% of those individuals without diabetes have heart disease. Likewise, the figure jumps to 19.8% for those with diabetes.

This difference is statistically significant and reveals that those with diabetes are over three times more likely to have heart disease as those without diabetes. In terms of description this reflects a significant variation between the two groups and as a measure of chronic condition there appears to be a significant impact on the general health of the population.

5.2.5 Heart Disease vs. Smoking Status

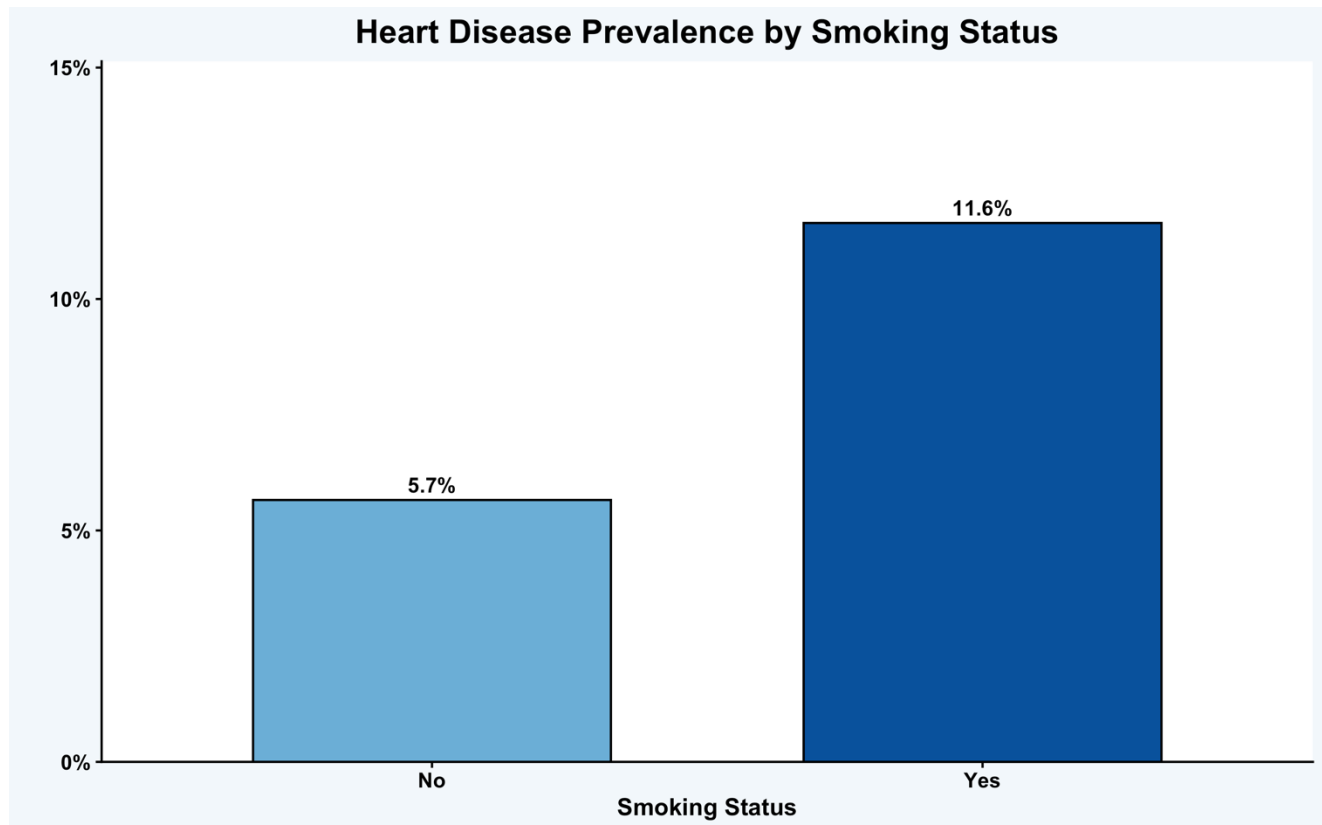


Figure 14

Prevalence of heart disease among those not smoking or having smoked: 5.7%.
Prevalence of heart disease among those who had smoked: 11.6%.

This difference presents a gap of almost 2:1, which reveals that among those who have smoked there are almost 2 times the proportion of people with heart disease than among those who do not. In a more descriptive sense this would point to an association between the two, given that a significant difference emerges in the "health for life" attribute.

6. Model Development

6.1 Modeling Approach

The objective of this project is to develop a predictive model capable of identifying individuals with heart disease using a large health dataset. A major challenge we encountered was an extreme imbalance for the target class (heart diseases). Over 91% of the respondents reported to have no heart disease with only 8% of the respondents who have a heart disease. Due to this challenge, machine learning models default to predicting the majority class which has no heart diseases. This leads to deceptively high accuracy with very low sensitivity, which poses a major challenge since health care professionals are more interested in detecting people with heart disease. To tackle this issue, we implemented a comprehensive modelling pipeline that included data preprocessing that included data preprocessing, SMOTE oversampling, Random Forest modeling using the efficient ranger package, and probability-based threshold tuning to improve the model's sensitivity to the minority class.

Most of the variables are categorical variables such as age, smoking status, exercise, so we employed dummy encoding allowing the model to interpret these variables numerically, to be able to apply SMOTE which relies on linear interpolation and distance metrics to create new datapoints. We then applied SMOTE (Synthetic Minority Oversampling Technique) to the training data. This is a technique that basically oversamples on the minority class while over sampling the majority class, this creates a more balanced exposure to both 'yes' and 'no'. SMOTE was only applied to the training set and the testing set remained untouched to reflect real world data, which will most likely be imbalanced for a rare disease like cardiovascular diseases.

Before, modeling we split the data into training and testing sets (80/20). Random forest was chosen as our first model due to its robustness to noise where we trained on 300 trees. When we first run the model with the default probability threshold of 0.50, There was a major issue, we achieved an overall high accuracy, however the model detected almost none of the true heart disease cases.

Sensitivity was 0.037 meaning the model missed nearly everyone who had a heart disease and specificity was high however that is misleading because the model was predicting the majority class and Kappa statistic showed almost no agreement beyond chance.

To address this, we tuned probability-based threshold tuning. Random forest default rule is that it predicts yes only when the probability is greater than 0.5, which is often strict for medical applications. For health care professionals, it is more important to detect potential cases even at the cost of some false positives, for that reason, we had to lower the threshold to allow the model to classify more individuals as 'yes' thus increasing sensitivity. Our threshold was ranging between 0.05 to 0.50 and we assessed the sensitivity, specificity, balanced accuracy, recall, F1 score at each level. The results showed a clear trade-off between sensitivity and specificity. (Refer in the appendix)

We found the optimal threshold to be 0.10 which produced the highest balanced accuracy of 0.748. At this level sensitivity also increased dramatically from the default 0.037 to 0.724, suggesting the model identified at least 72% of the true heart disease cases, specificity remained reasonably high as well which was 0.773 which means the model did a good job at identifying individuals without heart disease as well. In addition to the above metrics, we also evaluated precision, recall and F1 score to gain deeper understanding of the model's performance on the minority class. At the optimal probability threshold (0.10), the model achieved 0.2699, 0.4966 and 0.3498 respectively. This reflects a clear trade-off where improving sensitivity increases the number of false positives, which lowers precision. The primary goal of health care professionals is to detect true positives cases because undetected heart disease can silently lead to fatal outcomes whereby a false positive case leads to further testing which is far less harmful.

All models were trained using the reduced feature set obtained from the feature selection process described in Section 4.2. This approach improved model interpretability and reduced the impact of noisy or irrelevant predictors.

Building a robust pipeline for randomization and reproducibility across different ML models, the same modeling process was applied:

The initial dataset was randomly split into 80 % training and 20% test set using stratified sampling with respect to the target (heart disease) distribution, to mitigate the class imbalance (~8% positive case). Synthetic Minority Oversampling Technique (SMOTE) was only applied on the training set to generate synthetic examples to balance the positive class (~8%), avoiding information leakage.

All categorical covariates were transformed to one hot encoded vector, so that ML algorithms could use them. Continuous covariates were scaled (when required, e.g. for Support Vector Machines, etc.).

Model validation was carried out by all metrics specified below:

- Sensitivity (TPR)
- Specificity (TNR)
- Balanced accuracy
- Precision
- Recall
- F1 measure

Balanced accuracy was adopted as the metric of choice to assess the various classifiers, due to the class imbalance (roughly 8%).

Probability tune threshold was carried out for all trained classifiers, by applying a range of cut-offs from 0.05-0.50 in the model validation.

6.2 Random Forest

The Random Forest model was implemented using 300 trees (ntree = 300), which provided a balance between model stability and computational efficiency. At each split, a subset of features (mtry) was randomly selected, and the optimal value of mtry was determined through cross-validation.

The model minimizes classification error using Gini impurity as the splitting criterion. Feature importance was calculated using the mean decrease in Gini impurity, allowing identification of the most influential predictors in the model.

Hyperparameter tuning focused on:

- Number of trees (ntree)
- Number of variables tried at each split (mtry)
- Minimum node size

Cross-validation was used to evaluate model stability and prevent overfitting.

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6.3 XGBOOST

Model specifics and loss function

For the XGBoost model, the gradient boosting method was implemented with a logistic loss function (binary: logistic) suitable for a binary class decision.

The model would minimize the log-loss function iteratively through an ensemble of trees to optimize the predictive power of the last resulting tree.

Some of the hyperparameters such as:

- Learning rate (eta)
- Maximum tree depth (max_depth)
- Number of boosting rounds (nrounds)
- Minimum loss reduction (gamma)
- Subsample ratio (subsample)

were optimized using a cross-validation procedure for the best-balanced accuracy.

The concept of Gain (response to improved accuracy) was used to rank the importance of features.

Just like Random Forest, in XG boost we also used SMOTE on the training data to achieve a balanced dataset for both the 'Yes' and 'No'. We evaluated the XGBoost classifier across probability threshold of 0.05 to 0.50. As expected for rare diseases prediction task, lower threshold produces very high sensitivity, lower specificity, lower precision and higher recall. As we increase the probability threshold, we see sensitivity and recall going down gradually, and the other metrics increasing (You can refer in the appendix). At a threshold of 0.45, the model achieved a balanced accuracy of 0.757. At this threshold, precision was 0.193, recall 0.814, and the F1-score 0.312, with a sensitivity of 77% and specificity of 74% representing the best overall trade-off between identifying true heart disease cases and limiting false alarms.

This represents a fundamental trade-off in the medical field where lower thresholds maximize case detection at the expense of burdening clinicians with unnecessary follow-ups. While higher thresholds reduce false positives but at the cost of missing some true cases. Overall, XG boost classifier achieved a higher balanced accuracy as compared to Random Forest, as well as improved precision and F1 score suggesting that XGboost provides a better calibration between positive and negatives cases.

6.4 Support Vector Machine (SVM)

The SVM model was implemented using a radial basis function (RBF) kernel. The tuning parameters were set to $\sigma = 0.01$ and cost parameter $C = 1$, based on cross-validation results. Model performance was evaluated using 3-fold cross-validation.

Prior to model training, all predictors were centered and scaled, and categorical variables were converted into dummy variables to ensure compatibility with the SVM algorithm.

In contrast to the previous results, the SVM was used to predict whether the patient has heart disease. As before, the default probability cutoff point of 0.50 was used. The resulted model was the best overall accuracy obtained (around 89%) but the value of this is seriously questionable due to the class imbalance.

The model has quite a low sensitivity (around 35%) but very high specificity (around 94%), and therefore the balanced accuracy is not very high, indicating poor classification results in the test data for prediction of the minority class. It can be concluded that the trained SVM model showed quite a bias in using the majority class 'No' in prediction, which can be explained by the imbalanced data set. When comparing the two models, the logistic regression model performs better for this problem because it is more balanced and better at identifying heart disease cases. Even though the SVM model has higher accuracy, it does not perform well on the minority class, which is the more important outcome in this context.

There are also some limitations to consider. First, the dataset is highly imbalanced, and even with SMOTE, it is still challenging for the models to perform well. Second, the low precision for the positive class means that there are many false alarms. Third, the SVM model was trained on a smaller sample of the data due to computational limits, which may have affected its performance. Finally, accuracy alone is not a reliable metric in this case, so other measures like sensitivity and balanced accuracy are more important.

6.4.1 Enhancing SVM (RBF Kernel) Performance through Probability Threshold Tuning

In this analysis, a Support Vector Machine (SVM) model was trained on a sampled SMOTE dataset (50,000 observations) to address the class imbalance problem. When evaluated using the default probability threshold of 0.50, the model achieved a high accuracy of 89.13%. However, this result was misleading. The sensitivity was only 35%, meaning the model failed to detect many true heart disease cases. At the same time, specificity was very high (93.9%), indicating that the model was mainly predicting the majority class (“No”). The relatively low balanced accuracy (64.5%) and Kappa value further confirmed that the model performance was not well balanced.

To improve this, probability threshold tuning was applied. A range of thresholds from 0.05 to 0.50 was tested, and performance was evaluated using sensitivity, specificity, balanced accuracy, precision, recall, and F1 score. The results showed a clear trade-off: lower thresholds increased sensitivity but reduced specificity, while higher thresholds improved specificity at the cost of missing more true cases.

The optimal threshold was found to be 0.10, which produced the highest balanced accuracy of approximately 0.744. At this level, sensitivity increased significantly to 74.7%, meaning the model correctly identified most heart disease cases. Specificity remained reasonably high at 74.2%, showing that the model still performed well in identifying non-disease cases. Although precision was relatively low (about 20%), this is expected because improving sensitivity increases the number of false positives. The F1 score also improved, indicating a better balance between precision and recall.

Compared to the logistic regression model with SMOTE, the tuned SVM model shows a very similar performance. Both models achieve comparable balanced accuracy (around 0.74–0.75) and sensitivity, meaning they are equally effective at identifying heart disease cases. However, logistic regression is simpler and less computationally expensive, while SVM requires more tuning and processing time. Despite this, the SVM model still performs well after threshold adjustment and can be considered a strong alternative.

Overall, threshold tuning significantly improved the SVM model by reducing its bias toward the majority class. In a healthcare context, this trade-off is acceptable because detecting true cases is more important than avoiding false positives. Therefore, using a lower threshold such as 0.10 provides a more practical and meaningful approach for predicting heart disease.

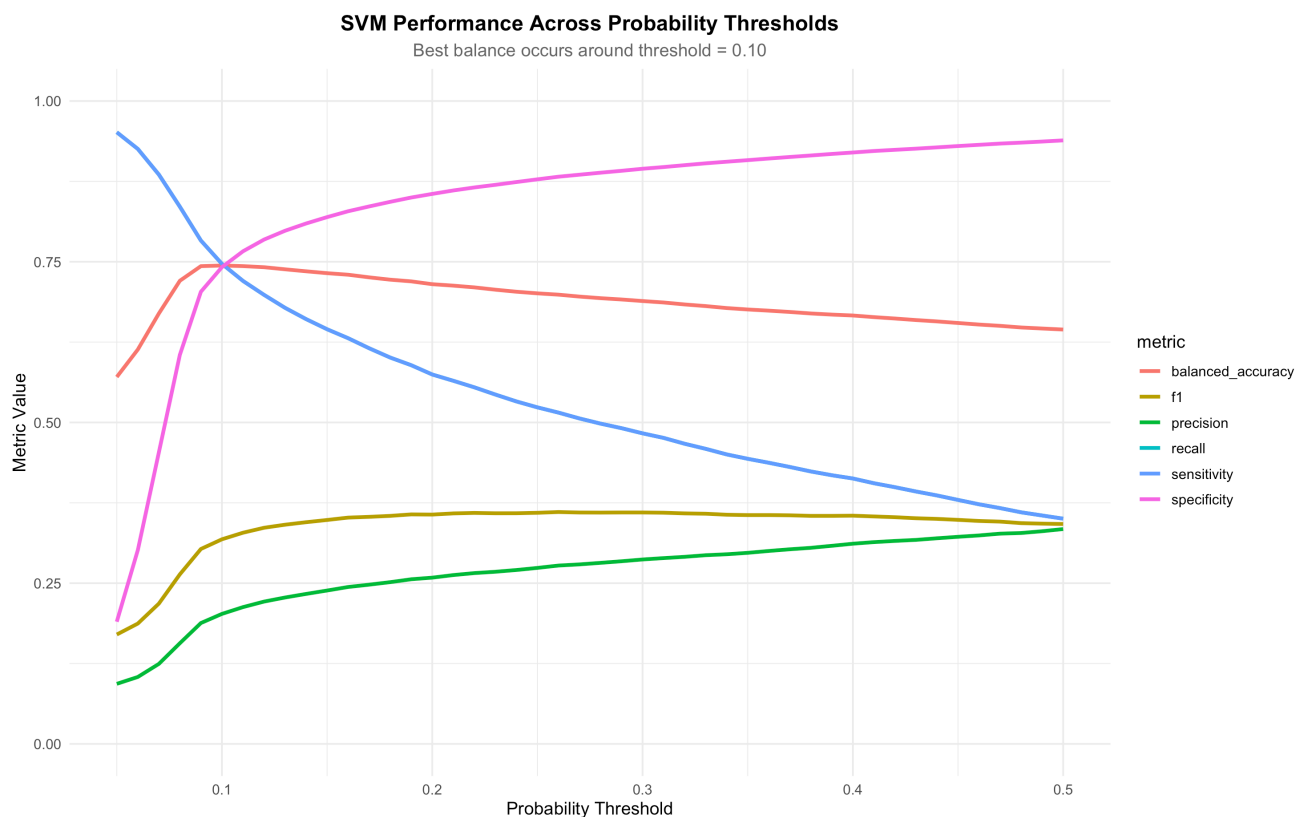


Figure 15

Figure 15 shows how the SVM model performance changes as we adjust the probability threshold. At lower thresholds, the model detects more heart disease cases (high sensitivity), but it also produces more false positives. As the threshold increases, the model becomes more conservative, improving specificity but missing more true cases. The best balance occurs around a threshold of 0.10, where the model achieves a good trade-off between detecting cases and avoiding errors.

This suggests that lowering the threshold improves the model's usefulness in identifying heart disease cases.

7. Limitations

Severe target class imbalance: Class imbalance was the biggest limitation affecting all models we used. The dataset was overwhelmingly imbalanced, with over 91% of the respondents having reported to have no heart disease, which means the models will most likely lean towards predicting 'NO' instead of yes, inflating the accuracy while hiding poor performance in the minority class. Even applying SMOTE on the training set, we still face the same challenge where the models achieved low sensitivity, for that reason we had to turn to probability-based threshold.

Synthetic Oversampling (SMOTE) has trade-offs: SMOTE generates synthetic points from the minority instances within the feature space that were occupied by the majority class.

This increases class overlap and can reduce classification performance.

All models we used are highly dependent on threshold tuning which means the default probability was useless.

Lack of clinical context: Most of our variables were binary categorical variables, it would have been nice to have more information about the heart disease such as the symptoms, duration, treatment history to enable the models to capture real world disease dynamics.

Self-reported health data is prone to bias- The dataset used is self-reported data which may potentially be biased. Bias such as recall bias, misreporting, or fear of reporting chronic illnesses, for example due to stigmatization, fear or embarrassment.

8. Conclusions

In this research study, cancer risk prediction models were built and tested on a way imbalanced data set. To deal with this imbalance, SMOTE and threshold tuning were implemented, achieving an increased sensitivity on the models.

Although accuracy scores ranged from 78 to 84%, they are not very informative in the case of an imbalanced data set. Balanced accuracy, sensitivity and specificity are more representative measures, with the last three measures showing that a significant number of positive cases could be identified through this approach by employing threshold tuning.

Moreover, variables such as general health, age, diabetes, arthritis and smoking history correlated positively with predisposition to having a heart disease.

All in all, results indicate that machine learning models, suitably tuned, could perform supporting tasks, such as early diagnosis. In the future, further model refinement (by including richer clinical datasets) and external validation should be pursued.

9. Recommendations

- **Integrate the model into routine clinical screening**

The three predictive models can be utilized for screening purposes in office visits, during diagnosis, to identify patients that have a high chance of being at risk of cardiovascular disease. This will lead to better management of patient potentially harboring the disease and consequently avoiding misdiagnosis. This also enables physicians to make better and more proactive decisions in the management of their patients and therefore adjust a treatment plan accordingly to a patient screening.

- **Prioritize high-risk groups for targeted intervention**

According to the analysis, those who have existing conditions of diabetes, arthritis and have smoked in past have much higher prevalence of heart disease. These clusters of individuals are to be focused for regular check-ups, targeted counselling's and preventive care interventions to make the most effective use of limited resources.

- **Expand access to preventive and lifestyle intervention programs**

High-risk patients should be referred to primary care and to community preventive care programs, like smoking cessation programs, physical activity programs and programs for chronic disease management.

- **Conduct external validation to assess generalizability**

To validate robustness of the models, we should look for test sets from different population or health care setting, as external validation is the best way to see the general effectiveness across different populations, health care systems and avoid potential overfitting data.

- **Enhance model performance with richer clinical data**

However, existing models are based only on demographic and self-reported variables. The inclusion of additional clinical variables—such as laboratory results, vital signs, and imaging data—would enable the model to capture more complex patterns and provide earlier insight into disease progression, thereby improving specificity.

10. References

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